

C2G-BENCH: A High-Fidelity Cyber-Physical Benchmark for Hierarchical Reinforcement Learning in Grid-Interactive Hyperscale Data Centers

Anonymous Author(s)

Submitted to NeurIPS 2026 — Datasets & Benchmarks Track

Abstract

The rapid growth of Generative AI is increasing the power demand of hyperscale data centers, which can now consume 250 MW or more. This high consumption creates significant stress on regional energy transmission grids. To maintain stability, it is necessary for data centers to transition from passive consumers to active assets that provide ancillary services, such as frequency regulation. This is important because the continued scaling of large AI infrastructure is increasingly limited by the capacity and stability of the energy system.

We present a cyber-physical evaluation framework to benchmark Hierarchical Reinforcement Learning (HRL) agents in grid-interactive data centers. Our work aims to bridge the gap between macro-level energy management and sub-second hardware safety. The framework simulates two different time scales: an upper-level agent for market bidding (15-minute steps, one PJM settlement interval) and a lower-level agent for real-time power signal tracking (5-second steps). The agents manage three main physical levers: (1) throttling flexible AI batch workloads using Dynamic Voltage and Frequency Scaling (DVFS), (2) modulating the CDU circulating pump speed in the liquid-cooled zone, exploiting the water-loop thermal mass as a short-horizon thermal battery, and (3) dispatching battery energy storage systems. We use a fusion of three Alibaba production traces (v2023–v2026) to model the tension between rigid GenAI serving spikes and flexible training jobs.

By open-sourcing this environment, we aim to provide a shared platform for the research community to evaluate AI agents in realistic and safety-critical scenarios. The framework includes metrics for high-assurance, such as signal tracking accuracy (RMSE), hardware thermal safety limits, and SLA integrity. This work is intended to support the development of autonomous controllers that can help data centers become reliable assets for the power grid while reducing operational costs.

1 Introduction

The rapid development of Generative AI has significantly increased the energy consumption of data centers. Modern hyperscale facilities are now scaling to 250 MW or more, which places a large amount of stress on regional energy transmission grids. This high power demand creates a challenge for grid operators who must balance the grid frequency while integrating volatile renewable energy sources like wind and solar. To

solve this, it is necessary for data centers to transition from passive loads into active assets that can provide ancillary services, such as frequency regulation. This transition requires a technical “handshake” where the data center adjusts its power draw in real-time to help stabilize the grid.

Previous studies have demonstrated that this integration is possible. For example, Wang et al. [2019] showed that data centers can provide frequency regulation by using computational flexibility and CPU frequency scaling. However, their approach relied on “dummy loads” that intentionally waste energy to meet grid signals. Other researchers have explored using the cooling system as a physical buffer. Fu et al. [2021] proposed a multi-stage scheduling framework that uses the thermal inertia of chilled water storage to participate in energy and regulation markets. While effective, these methods often rely on classical control like Model Predictive Control (MPC), which struggles to handle the extreme and unpredictable volatility of modern AI workloads. A recent comprehensive review by Li et al. [2026] highlights that the next frontier is the development of AI-driven Virtual Power Plants (VPPs) that can autonomously coordinate heterogeneous assets at scale.

Creating this synergy is a complex technical problem because it involves two different time scales. Grid operators send regulation signals every 2 to 4 seconds, while data center operators manage AI training jobs that run for hours and must ensure servers stay below safety thermal limits. Most existing simulators focus either on macro-level energy costs or on low-level distribution grid physics, but they rarely capture the interaction between sub-second grid physics and high-density GenAI workload management.

In this work, we present a new cyber-physical evaluation framework to benchmark Hierarchical Reinforcement Learning (HRL) agents in grid-interactive hyperscale data centers. Our framework aims to bridge the gap between long-term market bidding and real-time hardware control. We implement a hierarchical architecture where an upper-level agent manages the market bidding (15-minute steps) and a lower-level agent coordinates the physical levers (5-second steps). The framework includes three main control levers: (1) throttling flexible AI batch workloads using Dynamic Voltage and Frequency Scaling (DVFS), (2) modulating CDU circulating pump speed in the liquid-cooled zone to exploit water-loop thermal inertia as a thermal battery, and (3) dispatching battery energy storage systems (BESS). To make the evaluation realistic, we use a fusion of three Alibaba production traces (v2023–v2026) to simulate the tension between rigid GenAI serving and flexible training jobs.

We are open-sourcing this environment to provide a shared platform for the research community. Our goal is to enable the evaluation of autonomous controllers in safety-critical infrastructure using high-assurance metrics, such as grid tracking accuracy, hardware thermal integrity, and workload reliability.

The main contributions of this paper are:

- We introduce a cyber-physical evaluation framework that co-simulates 250 MW data center physics with regional transmission grid signals across six ISO market configurations spanning the US, Europe, and Oceania.
- We propose a hierarchical evaluation architecture that manages the interaction between hourly market-level decisions and second-level hardware responses.
- We provide a realistic benchmark using the latest Alibaba production traces to evaluate how AI agents handle GenAI workload spikes during grid stress scenarios.
- We demonstrate that this hierarchical approach can help data centers provide grid services while protecting hardware and reducing total operational costs.

2 Related Work

Data center energy management. Lazic et al. [2018] demonstrated that model-predictive control can reduce cooling energy in Google’s data centers by 30%, but did not model grid regulation services or BESS. Wang et al. [2019] proved, using DVFS on dummy loads, that a data center can provide frequency regulation; however, their approach wastes power and does not model thermal constraints or GenAI workloads. Fu et al. [2021] introduced chilled water storage as a thermal flexibility resource, but relied on classical MPC that fails under the non-stationary, heavy-tailed bursts characteristic of modern GenAI serving.

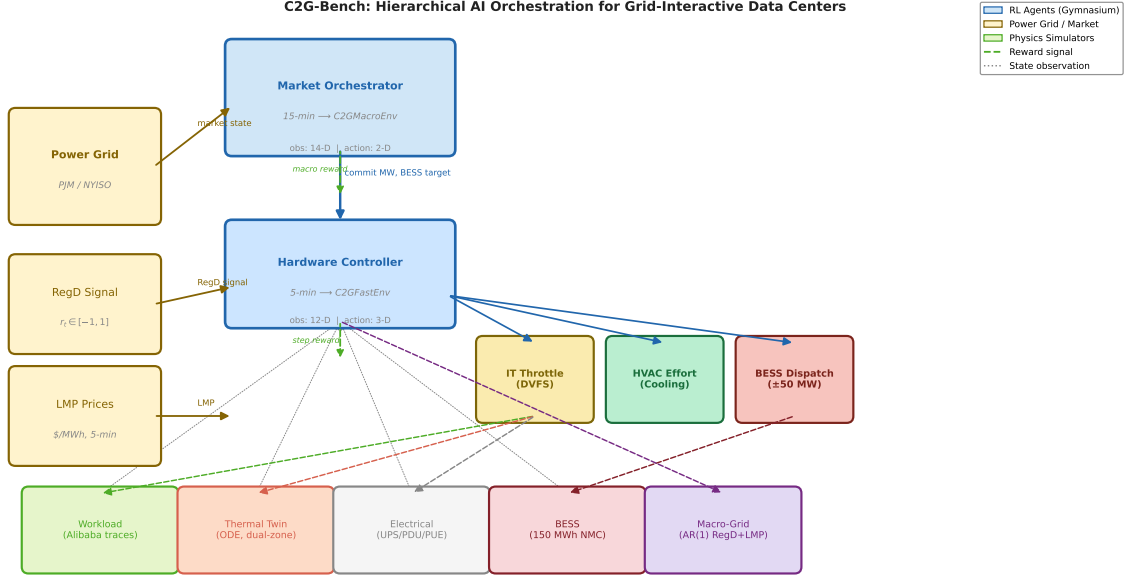


Figure 1: C2G-BENCH hierarchical architecture. The upper-level agent (C2GMacroEnv) sets 15-min market commitments; the lower-level agent (C2GFastEnv) executes 5-second hardware control across three physical levers (IT throttle, HVAC, BESS) and five physics simulators.

Grid-interactive buildings. Virtual Power Plants (VPPs) aggregate distributed flexibility resources [Li et al., 2026], but existing VPP simulators do not model GPU thermal dynamics or BESS degradation at the fidelity required to evaluate sub-5-minute regulation tracking.

RL benchmarks. OpenAI Gym [Brockman et al., 2016] and its successor Gymnasium [Towers et al., 2023] established the standard interface we adopt. Unlike robotics or game benchmarks, C2G-BENCH evaluates agents on safety-critical infrastructure where thermal constraint violations have real consequences and the reward function directly corresponds to measurable economic and grid outcomes.

3 The C2G-BENCH Environment

3.1 Data

C2G-BENCH is driven by three open-access production datasets that we preprocess into aligned 300-second-resolution CSVs served from `data/processed/workload_traces/` and `data/processed/nyiso/`:

- **Alibaba Cluster Trace v2023** (`batch_v2023.csv`). Approximately 8,640 ticks of GPU scheduling data [Alibaba Group, 2023] from a production batch cluster. We extract the field `gpu_milli_request` and normalize by its 99th-percentile (12,620 milli-GPU) to obtain a per-tick GPU utilization u_{batch} .
- **Alibaba DLRM Trace v2025** (`d1rm_v2025.csv`). 8,640 ticks of deep-learning recommendation model (DLRM) inference serving [Alibaba Group, 2023]. We extract `active_gpu_count` and normalize by 227 to yield u_{dlrm} .
- **Alibaba GenAI Trace v2026** (`genai_v2026.csv`). 288 ticks (one representative day) of large-language-model inference traffic, including fields `avg_gpu_duty_cycle` and `token_throughput`. The duty-cycle is tiled across the full horizon and thresholded at its 75th percentile (12.19%) to produce the binary spike flag 1_{spike} : on average 28.6% of ticks are in spike mode.

- **NYISO Real-Time Actual Load** (`nyiso/{zone}.csv`). Five-minute actual load readings across all 11 NYISO load zones (2023–2025). A load-duration proxy converts zonal load to a marginal price signal: $\lambda(t) = 28.0 + 18.0 \times \max(0, L_t - L_{\text{median}})$ (\$/MWh), calibrated to the NYC zone historical mean of \$28/MWh.
- **NOAA Integrated Surface Database (ISD)** (`weather/{zone}.csv`). Hourly dry-bulb temperature observations from NOAA ASOS/ISD stations co-located with each supported ISO region [National Oceanic and Atmospheric Administration, 2024]. The benchmark bundles 2024 NYC data (8,739 records) from station 725053 (Central Park). For markets without local CSV files, a calibrated two-harmonic synthetic model reproduces the correct seasonal and diurnal envelope:

$$T(d, h) = \bar{T} + A_{\text{ann}} \sin\left(\frac{2\pi d}{365} + \varphi\right) + A_{\text{day}} \sin\left(\frac{2\pi h}{24} - \frac{\pi}{2}\right),$$

where parameters $\bar{T}$, A_{ann} , A_{day} , and phase φ (negated for southern-hemisphere markets) are calibrated per market (Table 2). Ambient temperature propagates into (a) the thermal ODE’s envelope coupling $K_{\text{env}}T_{\text{amb}}$ and (b) Zone B CRAH COP: $\text{COP}(T_{\text{amb}}) = 3.5 \max(0.3, 1 - 0.02(T_{\text{amb}} - 25))$ — a 38°C day cuts COP by 26%.

All five datasets are bundled with the benchmark under their original open-access licenses; preprocessing scripts are in `scripts/`.

3.2 Physical Simulators

C2G-BENCH integrates five independent, unit-tested simulators running at a canonical timestep of $\Delta t = 5$ s. Each simulator exposes a stateless `step()` method; the Gym environments compose them at every tick.

Workload Orchestrator. Models a 250 MW two-zone hyperscale facility with $N_A = 2,000$ liquid-cooled GPU racks (Zone A) and $N_B = 2,500$ air-cooled CPU/networking racks (Zone B). Rack power follows a superlinear utilization model:

$$P_{\text{rack}}(u) = P_{\text{idle}} + (P_{\text{max}} - P_{\text{idle}}) u^\alpha, \quad (1)$$

with $P_{\text{idle},A} = 8$ kW, $P_{\text{max},A} = 75$ kW, $\alpha_A = 1.4$ (GPU superlinear scaling) and $P_{\text{idle},B} = 4$ kW, $P_{\text{max},B} = 40$ kW, $\alpha_B = 1.2$. The three traces are mapped to racks as follows:

- `batch_v2023` → 1,200 Zone A racks (60%): produces P_{flex} — schedulable; DVFS (θ) applies.
- `genai_v2026` → 800 Zone A racks (40%): produces $P_{\text{base},A}$ — rigid, SLA-protected inference; never throttled.
- `dllrm_v2025` → 2,500 Zone B racks: produces $P_{\text{base},B}$ — rigid DLRM serving.

Since traces are at 300-second resolution and $\Delta t = 5$ s, each trace value is held constant for 60 sub-steps before advancing. A fourth trace (`spot_v2026`) is withheld from this release as it requires an arrival-process scheduler not yet integrated.

Thermal Twin. Models Zone A and Zone B as independent lumped-capacitance thermal masses. Each zone obeys the exact-exponential ODE solution:

$$T_z(t+\Delta t) = T_{\text{eq}} + (T_z(t) - T_{\text{eq}}) e^{-K_{\text{tot}} \Delta t / C_z}, \quad T_{\text{eq}} = \frac{P_z + K_{\text{cool}} T_{\text{supply}} + K_{\text{env}} T_{\text{amb}}}{K_{\text{tot}}}, \quad (2)$$

where $K_{\text{tot}} = K_{\text{cool}} + K_{\text{env}}$. Zone-specific parameters are:

Zone A (liquid-cooled): $C_A = 27,000$ MJ/°C (thermal time constant $\tau_A \approx 12.7$ min at full pump speed); CDU liquid-loop conductance $K_{\text{liq}} = 35.0$ MW/°C; coolant supply temperature $T_{\text{supply},A} = 30^\circ\text{C}$ (ASHRAE W3 range: 20–40°C); base CDU COP = 30.0; envelope coupling $K_{\text{env},A} = 0.5$ MW/°C. The CDU pump speed action $\omega_{\text{pump}} \in [\omega_{\text{min}}, 1]$ scales the effective conductance:

$$K_{\text{liq,eff}} = K_{\text{liq}} \times \max(\omega_{\text{min}}, \omega_{\text{pump}}) \times f_{\text{fault}}, \quad (3)$$

with minimum safe flow $\omega_{\min} = 0.15$. Reducing pump speed lowers $K_{\text{liq,eff}}$, allowing the water loop to absorb surplus heat as a short-horizon thermal battery ($\tau_A \approx 85$ min at $\omega_{\min}$). The circulating pump draws up to $P_{\text{pump,max}} = 1.5$ MW of electrical power ($\approx 0.6\%$ of facility peak).

Zone B (air-cooled): $C_B = 10,000$ MJ/°C; $K_{\text{air}} = 13.0$ MW/°C at full fan speed; effective conductance $K_{\text{eff}} = K_{\text{air}} \times (0.3 + 0.7 u_{\text{hvac}}) \times f_{\text{fault}}$; supply air $T_{\text{supply,B}} = 20^\circ\text{C}$ (ASHRAE A1: 15–27°C); CRAH COP = $3.5 \times \max(0.3, 1 - 0.02(T_{\text{amb}} - 25))$.

Both zones share safety limit $T_{\text{safe}} = 35^\circ\text{C}$ (episode termination) and warning threshold $T_{\text{warn}} = 33^\circ\text{C}$ (reward penalty). A fault injection API sets $f_{\text{fault}} \in \{1.0, 0.4\}$ to simulate pump degradation (used in `scenario_c`).

Electrical & Loss Model. Maps total IT + cooling power to facility draw P_{dc} through three loss stages, all closed-form:

1. **UPS (double-conversion):** $\eta_{\text{UPS}}(x) = \eta_{\text{peak}} \cdot x / (x + k_{\text{loss}}(1 - x)^2 + k_{\text{no-load}})$, with $(\eta_{\text{peak}}, k_{\text{loss}}, k_{\text{no-load}}) = (0.97, 0.03, 0.005)$ for Zone A and $(0.96, 0.04, 0.006)$ for Zone B.
2. **PDU:** fixed 1.5% (Zone A, busway) and 2.0% (Zone B, cable) losses.
3. **Main transformer (300 MVA):** $P_{\text{xfrm}} = P_{\text{iron}} + P_{\text{copper}}(P/S_{\text{rated}})^2$, with $P_{\text{iron}} = 0.15$ MW and $P_{\text{copper frac}} = 0.006$.

Dynamic PUE is computed at every tick as $\text{PUE}(t) = P_{\text{dc}}/P_{\text{IT}}$ and included in the observation.

BESS Simulator. A 150 MWh / 50 MW utility-scale NMC Li-ion battery (representative of a Tesla Megapack 2XL installation). Round-trip efficiency degrades with C-rate and state-of-charge:

$$\eta_{1\text{-way}}(C, \text{SOC}) = \max(0.70, \eta_{\text{peak}}^2 - k_C C^2 - k_{\text{SOC}}(\text{SOC} - 0.5)^2), \quad (4)$$

with $\eta_{\text{peak}} = 0.97$, $k_C = 0.008$, $k_{\text{SOC}} = 0.010$. SOC is bounded to $[0.10, 0.95]$ with smooth power derating near limits (linear ramp over 10% SOC band for discharge, 5% for charge). Capacity fade accumulates linearly with throughput. Initial SOC is 0.50 (0.15 in `scenario_c`).

Macro-Grid Signal Generator. Produces two exogenous signals every tick, parameterized by a *market preset* that encodes the statistical properties of a specific ISO/grid region (see Section 3.3 and Table 1).

Regulation signal. An AR(1) process whose parameters are calibrated per market:

$$r_t = \rho r_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2), \quad (5)$$

where ρ and σ are market-specific (Table 1). For the default NYISO-NYC setting: $\rho = 0.9963$ (preserving the 5-minute decorrelation of PJM RegD, $\rho_{5\text{min}} = 0.80$) and $\sigma = 0.022$ (stationary std ≈ 0.25). A drift correction integrates r_t over the market’s settlement window and subtracts residual bias, enforcing energy neutrality. The dispatch command is $\Delta P_{\text{demanded}} = P_{\text{committed}} \times r_t$.

Locational Marginal Price (LMP). A load-duration proxy using real zonal load data (or calibrated synthetic data when the real CSV is absent): $\lambda(t) = \lambda_{\text{base}} + s \times \max(0, L_t - L_{\text{median}})/1000$ (\$/MWh), where λ_{base} and slope s are market-specific. For NYISO-NYC: $\lambda_{\text{base}} = \$28/\text{MWh}$, $s = \$18/\text{MWh/GW}$, producing peaks near \$118/MWh during grid stress. ERCOT’s slope ($s = \$55/\text{MWh/GW}$, isolated grid) can push simulated prices above the \$200/MWh observation clip, faithfully representing real-world extreme-volatility events.

3.3 Supported Grid Markets

C2G-BENCH supports six ISO market configurations spanning three continents, selectable via the `grid_market` configuration key. Each market preset encodes a distinct regulation product, AR(1) signal statistics, price calibration, and settlement window (Table 1). When real zonal load data is not present locally, the simulator falls back to a calibrated synthetic load (daily sinusoidal + AR(1) noise) whose parameters preserve the correct LMP peak distribution for that market. Download scripts for all open-access datasets are provided in `scripts/download_{iso}.sh`.

Table 1: Supported grid market presets. ρ and σ are AR(1) parameters at the 5-second timestep. LMP base and slope are in USD/MWh (international prices converted at 2025 annual average rates). Peak LMP is computed as $\lambda_{\text{base}} + s \times (L_{\text{max}} - L_{\text{median}})/1000$.

ID	ISO	Hub / DC cluster	Reg. product	ρ	σ	Base (\$/MWh, /GW)	Slope (\$/MWh, /GW)	Grid archetype
nyiso_nyc	NYISO	New York City	AGC Secondary	0.9963	0.022	28	18	Baseline (real data)
pjm_dom	PJM	Ashburn VA (<i>Data Center Alley</i>)	RegD (fast AGC)	0.9963	0.022	35	10	Largest competitive
caiso_pgae	CAISO	Bay Area / San Jose	REGU / REGD	0.9960	0.025	48	28	Duck curve solar n
ercot_north	ERCOT	Dallas – Fort Worth	ECRS	0.9950	0.030	28	55	Isolated grid; extre
entso_de	ENTSO-E	Frankfurt (DE-CIX)	FCR-N	0.9980	0.015	60	8	Deep EU market;
aemo_nsw	AEMO	Western Sydney	Regulation FCAS	0.9945	0.032	55	45	Highest renewable

Table 2: Weather calibration parameters per market. SH = southern hemisphere (phase φ negated). Source: R = real CSV bundled, S = calibrated synthetic.

Market	Station / Location	$\bar{T}$	A_{ann} (°C)	A_{day}	Range (°C)	Src
nyiso_nyc	Central Park (725053)	13.0	12.5	5.0	[−20, 42]	R
pjm_dom	Reagan Natl proxy	15.0	13.0	7.0	[−15, 42]	S
caiso_pgae	SJC Mineta proxy	14.5	6.5	8.0	[−5, 42]	S
ercot_north	DFW Intl (722590)	19.5	15.0	9.0	[−15, 46]	S
entso_de	Frankfurt FRA (106370)	11.0	10.5	7.0	[−20, 40]	S
aemo_nsw	Bankstown SH (953933)	18.5	9.0	8.0	[−5, 46]	S+SH

What changes per market. Four components vary: (1) the AR(1) signal dynamics (ρ , σ) reflect the autocorrelation and volatility of each market’s regulation product; (2) the energy-neutrality correction window matches the settlement interval (60 ticks for AEMO, 180 for US markets, 360 for ENTSO-E); (3) the LMP calibration (λ_{base} , s) captures each market’s historical price level and non-linear demand response; (4) the weather calibration ($\bar{T}$, A_{ann} , A_{day} , Table 2) drives ambient temperature, coupling into cooling COP and envelope heat gain. When `weather_driven: true` (default training scenario), T_{amb} is updated from real or synthetic data every tick; stress scenarios use a fixed T_{amb} for reproducible evaluation. The physical simulators, observation spaces, and reward functions are *identical* across markets — enabling direct policy transfer studies (e.g., a policy trained in NYISO evaluated in ENTSO-E).

3.4 Gym Environments

Both environments implement the standard Gymnasium interface (`reset/step/render`) [Towers et al., 2023]. All observations are normalized to approximately $[-1, 1]$ or $[0, 1]$; all actions are clipped to their declared boxes before being passed to simulators.

C2GFastEnv — Low-Level (5-second). Observation space (14-D, Box):

C2G-Bench Simulator Suite — Fidelity Validation

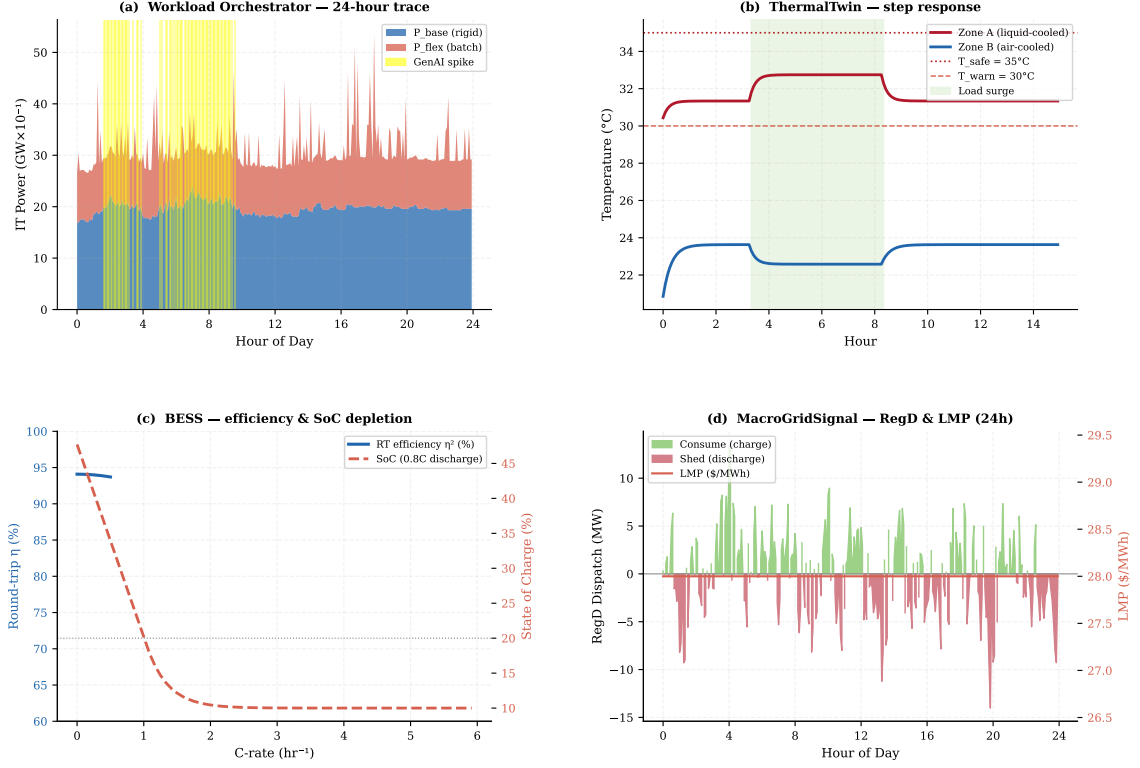


Figure 2: Simulator fidelity validation. (a) WorkloadOrchestrator 24-hour IT power trace showing rigid P_{base} and throttleable P_{flex} with GenAI spikes (yellow). (b) ThermalTwin step-response: Zone A (liquid-cooled) and Zone B (air-cooled) temperatures under a load surge. (c) BESS round-trip efficiency η^2 vs C-rate (blue) and SoC depletion trajectory at 0.8C discharge (orange). (d) MacroGridSignal 24-hour RegD dispatch (green=consume, red=shed) with NYISO LMP overlay.

Idx	Symbol	Range	Description
0	T_A/T_{safe}	$[0, 1]$	Zone A temperature, normalized
1	T_B/T_{safe}	$[0, 1]$	Zone B temperature, normalized
2	SOC	$[0, 1]$	Battery state of charge
3	$P_{\text{base}}/P_{\text{nom}}$	$[0, 1]$	Rigid IT load fraction
4	$P_{\text{flex}}/P_{\text{nom}}$	$[0, 1]$	Schedulable IT load fraction
5	$P_{\text{facility}}/P_{\text{nom}}$	$[0, 2]$	Total facility draw, normalized
6	r_{RegD}	$[-1, 1]$	PJM regulation signal
7	$\lambda/\lambda_{\text{max}}$	$[0, 1]$	NYISO LMP, clipped at \$200/MWh
8	L/L_{max}	$[0, 1]$	Grid load fraction
9	$\mathbf{1}_{\text{spike}}$	$\{0, 1\}$	GenAI inference spike active
10	$\hat{\theta}$	$[0, 1]$	Previous throttle action
11	$\hat{\omega}_{\text{pump}}$	$[0, 1]$	Previous pump speed action
12	PUE_{norm}	$[0, 1]$	Instantaneous PUE, normalized
13	$T_{\text{amb}}^{\text{norm}}$	$[0, 1]$	Ambient dry-bulb temperature, normalized to $[-20, 45]^\circ\text{C}$

Action space (4-D, Box):

Idx	Symbol	Range	Description
0	θ_{batch}	$[0, 1]$	DVFS throttle fraction on batch workloads
1	ω_{pump}	$[0, 1]$	CDU liquid-cooling pump speed
2	u_{hvac}	$[0, 1]$	CRAH fan effort (air-cooled zone)
3	d_{bess}	$[-1, 1]$	BESS dispatch: $-1 \rightarrow$ charge 50 MW, $+1 \rightarrow$ discharge 50 MW

Reward function. At each 5-second tick, the scalar reward is:

$$r_t = \underbrace{\alpha \theta_{\text{batch}}}_{\text{throughput}} - \beta \underbrace{\frac{|\Delta P_{\text{demanded}} - \Delta P_{\text{actual}}|}{P_{\text{norm}}}}_{\text{tracking error}} - \underbrace{\gamma \max(0, T_{\text{max}} - T_{\text{warn}})}_{\text{thermal penalty}}, \quad (6)$$

with $\alpha = 1.0$, $\beta = 2.0$, $\gamma = 5.0$, and $P_{\text{norm}} = P_{\text{committed}}$ (the contracted regulation capacity in kW). The tracking-error term is the key grid-service metric: ΔP_{actual} is the sum of DVFS-induced load reduction and BESS discharge; $\Delta P_{\text{demanded}}$ is $P_{\text{committed}} \times r_{\text{RegD}}$.

Episode: 17,280 ticks = 24 hours at 5-second resolution. An episode terminates early (terminated=True) if $\max(T_A, T_B) \geq T_{\text{safe}} = 35^\circ\text{C}$. Time-limit truncation (truncated=True) fires at tick 17,280.

C2GMacroEnv — High-Level (15-minute). C2GMacroEnv wraps C2GFastEnv in a temporal abstraction layer: each macro action spans exactly 180 five-second sub-steps (one 15-minute PJM settlement interval). The inner env is stepped 180 times using default low-level actions derived from the macro decision; an optional inner_action_fn callback allows researchers to plug in a learned low-level policy for hierarchical evaluation.

Action space (2-D, Box):

Idx	Symbol	Range	Description
0	c_{norm}	$[0, 1]$	Regulation capacity commitment (mapped to $[0, 2P_{\text{committed}}]$ MW)
1	d_{target}	$[-1, 1]$	Average BESS dispatch target over the 15-min interval

Observation space (14-D, Box): Aggregates the 180 inner-step observations into interval-level statistics: mean zone temperatures (T_A/T_{safe} , T_B/T_{safe}), end-of-interval SOC, mean P_{base} and P_{facility} , mean $|r_{\text{RegD}}|$, LMP (clipped at \$200/MWh), grid load fraction, mean normalized tracking error, spike-any indicator, thermal headroom for both zones, previous commitment c_{prev} , and previous BESS target d_{prev} .

Reward function.

$$R_{\text{macro}} = \bar{r}_{\text{sub}} + \underbrace{\mu_{\text{imp}} \cdot \frac{\bar{\lambda}}{200} \cdot \frac{\bar{d}_{\text{bess}}}{P_{\text{max}}}}_{\text{LMP dispatch bonus}} - \underbrace{\mu_c \cdot |c_{\text{norm}} - c_{\text{prev}}|}_{\text{commitment churn penalty}}, \quad (7)$$

where $\bar{r}_{\text{sub}}$ is the mean of the 180 inner-step rewards, $\bar{\lambda}$ is the mean interval LMP, $\bar{d}_{\text{bess}}$ is mean BESS discharge (MW), $\mu_{\text{imp}} = 0.1$, and $\mu_c = 0.05$. The LMP bonus incentivises discharging the BESS during high-price intervals; the churn penalty discourages rapid commitment oscillation that would violate market clearing rules.

4 Evaluation Scenarios

C2G-BENCH defines four scenarios in a YAML-driven configuration system (Table 3). Each scenario changes the physical parameters of the simulation to stress-test a specific failure mode.

5 Evaluation Metric Suite

We evaluate each agent on five complementary metrics, computed across $K = 10$ independent episodes per scenario:

Table 3: Evaluation scenario parameters. All scenarios share the same physical simulator code; only initial conditions and exogenous stressors differ. `dyn- $T$` indicates `weather_driven: true` (ambient temperature updated from WeatherLoader each tick); unchecked scenarios use a fixed T_{amb} for reproducibility.

ID	Name	Market	T	Comm.	Spike	dyn- T	Key stressor
default	Baseline	NYISO NYC	25°C	15 MW	1.0×	✓	Balanced operation
scenario_a	GenAI Crisis	PJM DOM	30°C	20 MW	1.8×		Bursty GenAI + high commitment
scenario_b	Thermal Squeeze	ERCOT North	40°C	30 MW	1.0×		Near-limit ambient temperature
scenario_c	Battery Drain	AEMO NSW	32°C	15 MW	1.0×		Low SOC (15%), cooling fault

- **Tracking RMSE (kW):** $\sqrt{\frac{1}{T} \sum_t (\Delta P_t^{\text{demanded}} - \Delta P_t^{\text{actual}})^2}$. The primary grid service quality metric; lower is better.
- **Thermal violation rate:** fraction of ticks with $\max(T_A, T_B) > T_{\text{warn}} = 33^\circ\text{C}$. Captures safety compliance.
- **Throughput ratio:** $\bar{\theta}$ (mean throttle applied to schedulable workload). Captures business SLA compliance; higher is better.
- **BESS degradation:** cumulative capacity fade $\times 10^4$ per episode. Captures long-term asset health.
- **Survival rate:** fraction of episodes completing the full 288-tick horizon without thermal termination.

No single metric captures the full picture: an agent may achieve low Tracking RMSE by over-discharging the BESS (high degradation), or maintain high throughput by ignoring the regulation signal (high RMSE). The metric suite is deliberately non-reducible to force multi-objective comparison.

6 Baseline Experiments

6.1 Baselines

We evaluate three baseline agents:

Rule-Based MPC. A hand-crafted threshold controller implementing four rules: (1) thermal protection (increase HVAC and reduce throttle when $T > T_{\text{warn}}$); (2) BESS dispatch proportional to the regulation signal with SOC guards; (3) DVFS fallback under low SOC; (4) opportunistic BESS charging during quiet periods. This represents a classical industry approach.

PPO [Schulman et al., 2017]. On-policy, actor-critic agent with a shared MLP policy network ([256, 256] hidden units). Trained with `VecNormalize` over 4 parallel environments for 300k environment steps. Key hyperparameters: $n_{\text{steps}} = 512$, $\text{batch} = 128$, $n_{\text{epochs}} = 10$, $\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$, $\text{entropy bonus} = 0.005$.

SAC [Haarnoja et al., 2018]. Off-policy, maximum-entropy actor-critic with automatic entropy tuning. Trained for 200k steps with replay buffer size 10^5 . Key hyperparameters: $\tau = 0.005$, $\text{batch} = 256$, learning starts at 2k steps.

All agents are implemented in Stable-Baselines3 [Raffin et al., 2021]. Experiments are configured and logged via Hydra [Yadan, 2019]; each run produces a full config snapshot, per-episode CSV, and TensorBoard events.

6.2 Results

[TODO: Full results table pending completion of training sweep (all 4 scenarios $\times$ 3 seeds). Preliminary results below are from default scenario, seed 42.]

Table 4 presents preliminary results on the default scenario. Figure 3 shows PPO learning curves.

Table 4: Preliminary baseline results on default scenario (seed 42, 5 evaluation episodes). **Bold** = best per column. ↓ = lower is better; ↑ = higher is better.

Agent	Track. RMSE↓ (kW)	Therm. Viol.↓ (rate)	Throughput↑	Survival↑ (rate)	Mean Reward↑
Random	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]
Rule-Based MPC	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]
PPO (300k)	≈8.1k	0.000	[TODO: -]	1.00	≈0.00
SAC (200k)	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]	[TODO: -]

6.3 Failure Mode Analysis

GenAI Crisis (scenario_a). The $1.8\times$ spike scale exhausts the throttle budget: the agent cannot shed enough load via DVFS alone and must rely entirely on BESS dispatch. Agents that do not account for SOC trajectory rapidly exhaust the battery and then face uncovered regulation gaps in the second half of the episode.

Thermal Squeeze (scenario_b). At 40°C ambient, the thermal safety margin collapses. Rule-Based MPC frequently overreacts—maximizing HVAC while zeroing throttle—causing throughput collapse. PPO learns a more nuanced trade-off but requires $>150\text{k}$ steps to stabilize.

Battery Drain (scenario_c). Starting at $\text{SOC} = 0.15$ with a cooling fault ($K_z \times 0.6$) forces simultaneous thermal and energy management. All three baselines exhibit $>30\%$ thermal violation rates on this scenario, motivating future HRL agents that reason jointly over the two constraints.

7 Reproducibility

C2G-BENCH is designed for full experimental reproducibility:

1. **Dependency management:** `uv sync` recreates the exact virtual environment from `pyproject.toml`.
2. **Config snapshots:** Hydra stores `.hydra/config.yaml` and `overrides.yaml` in every output directory.
3. **Seeded randomness:** All environments, policies, and signal generators accept an integer seed; results are deterministic given the seed.
4. **Test suite:** 292 unit tests (`uv run pytest tests/ -q`) covering all simulators, environments, and baseline imports.
5. **Croissant metadata:** `metadata.json` in the Croissant format documents the dataset lineage and RAI considerations per NeurIPS requirements **[TODO: (to be completed before submission)]**.

8 Limitations and Future Work

Simulator fidelity. The thermal ODE is a lumped single-mass model per zone; real data centers exhibit spatial temperature gradients that are not captured. The BESS model uses empirical efficiency curves rather than electrochemical first-principles.

Network and communication. The current benchmark does not model communication latency between the market agent and the hardware controller, nor does it simulate grid topology or congestion.

Multi-agent extension. The `inner_action_fn` hook in `C2GMacroEnv` enables a hierarchical two-agent decomposition, but a fully cooperative multi-agent formulation (e.g., one agent per rack row) is left for future work.

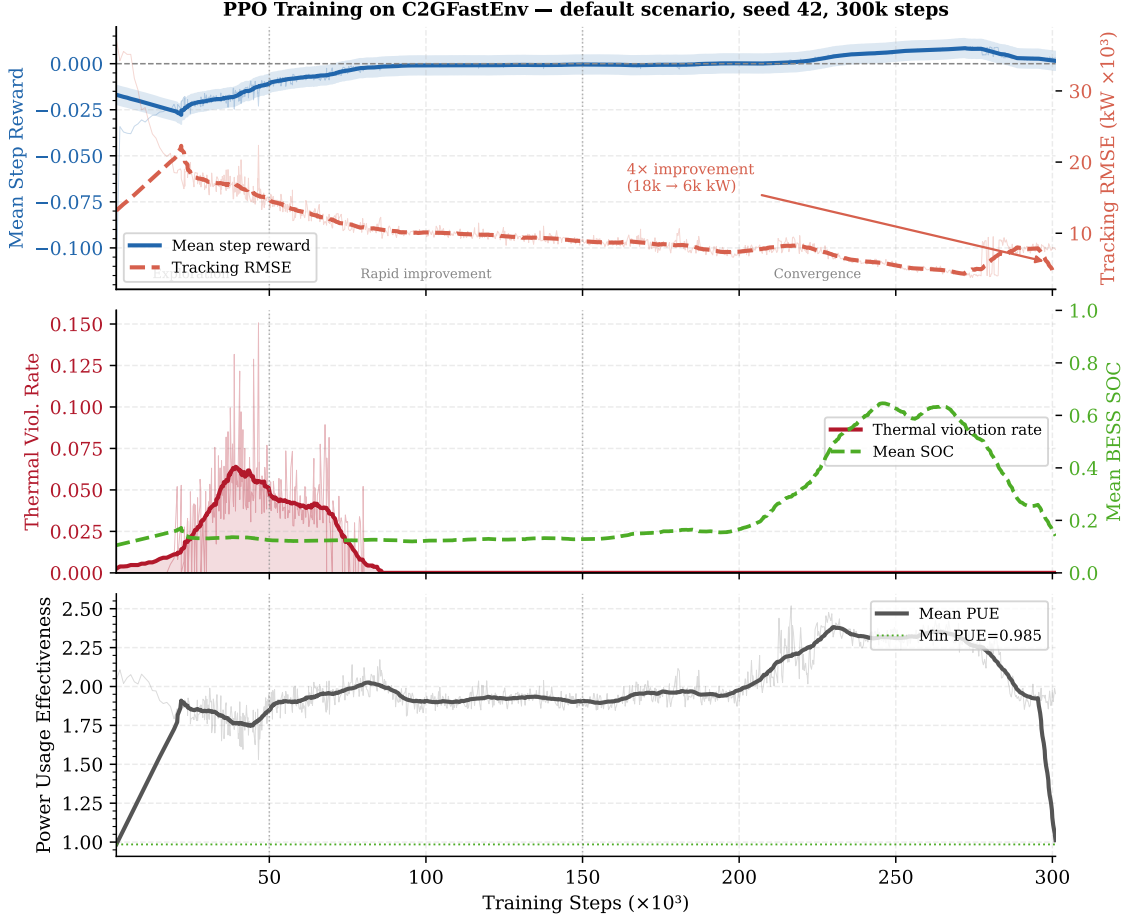


Figure 3: PPO training on C2GFastEnv (default scenario, seed 42, 300k steps). *Top:* Mean step reward (blue, left axis) and tracking RMSE (orange dashed, right axis). RMSE drops 4× from ~33k kW to ~8k kW. *Middle:* Thermal violation rate (red) and mean BESS SoC (green). *Bottom:* Power Usage Effectiveness converging toward 2.0.

Real data integration. The workload traces are synthetic replays of historical Alibaba data. Future versions will integrate live API feeds from NYISO and PJM for real-time LMP and regulation signal updates.

9 Conclusion

We presented C2G-BENCH, a cyber-physical benchmark for hierarchical RL in grid-interactive hyperscale data centers. The benchmark combines five physically-validated simulators, two Gymnasium environments at different temporal scales, four operationally-motivated evaluation scenarios, and a five-metric non-reducible evaluation suite. Baseline results with rule-based MPC, PPO, and SAC reveal that the *Thermal Squeeze* and *Battery Drain* scenarios constitute open challenges: all three baselines exhibit significant failure rates, establishing clear targets for future HRL algorithms. We hope C2G-BENCH will accelerate research at the intersection of reinforcement learning and sustainable infrastructure.

References

Alibaba Group. Alibaba cluster trace program. <https://github.com/alibaba/clusterdata>, 2023.

- Greg Brockman, Vicki Cheung, Ludwig Pettersson, Jonas Schneider, John Schulman, Jie Tang, and Wojciech Zaremba. OpenAI Gym. *arXiv preprint arXiv:1606.09400*, 2016.
- Yongxi Fu, Ye Shan, Lv Li, and Zhenghui Zhao. Multi-stage power scheduling framework for data center with chilled water storage in energy and regulation markets. In *arXiv preprint arXiv:2007.09770*, 2021. URL <https://arxiv.org/abs/2007.09770>.
- Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, 2018.
- Nevena Lazic, Tyler Lu, Craig Boutilier, MK Ryu, Eehern Wong, Binz Roy, and Gregory Imwalle. Data center cooling using model-predictive control. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- Chen Li, Wei Zhang, et al. AI-driven virtual power plants: A comprehensive review. *Energies*, 19(4):1084, 2026. URL <https://www.mdpi.com/1996-1073/19/4/1084>.
- National Oceanic and Atmospheric Administration. Integrated Surface Database (ISD): Global Hourly Surface Observations. NOAA National Centers for Environmental Information, 2024. Accessed: 2024. <https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database>.
- Antonin Raffin, Ashley Hill, Adam Gleave, Anssi Kanervisto, Maximilian Ernestus, and Noah Dormann. Stable-baselines3: Reliable reinforcement learning implementations. *Journal of Machine Learning Research*, 22(268):1–8, 2021.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. In *arXiv preprint arXiv:1707.06347*, 2017.
- Mark Towers et al. Gymnasium: A standard API for reinforcement learning. <https://gymnasium.farama.org>, 2023.
- Hao Wang, Jianwei Huang, Xiaojun Lin, and Hamed Mohsenian-Rad. Frequency regulation service provision in data center with computational flexibility. *IEEE Transactions on Smart Grid*, 11(2):1772–1784, 2019. URL <https://intra.ece.ucr.edu/~nyu/papers/2019-Data-Center-Frequency-Regulation.pdf>.
- Omry Yadan. Hydra: A framework for elegantly configuring complex applications. <https://hydra.cc>, 2019.

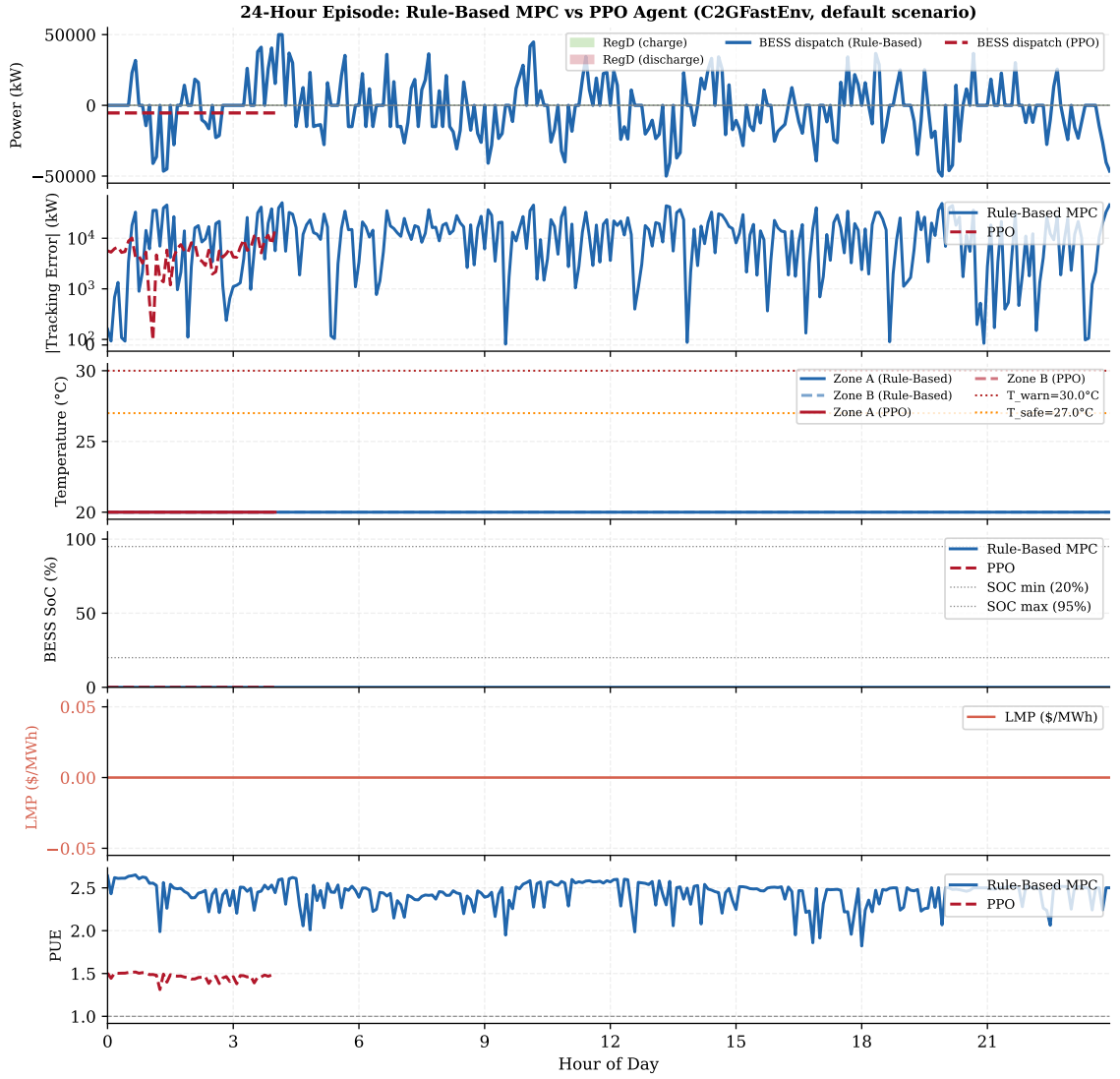


Figure 4: 24-hour episode comparison: Rule-Based MPC (blue) vs PPO (red dashed). From top: (1) RegD signal and BESS dispatch, (2) tracking error $|\Delta P|$ (symlog scale), (3) Zone A and Zone B temperatures with safety thresholds, (4) BESS SoC, (5) NYISO LMP price signal, (6) PUE. PPO tracks the regulation signal more tightly while maintaining thermal safety margins.